Real-Time Face Mask Detection Using Transfer Learning and OpenCV

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Introduction

Background & Motivation

The COVID-19 pandemic fundamentally changed public health practices worldwide. Face masks became critical preventive measures in crowded environments and healthcare facilities.

Problem: Manual enforcement of mask compliance faces multiple challenges:

- **Inefficient:** Requires continuous human supervision
- **Error-prone:** Subject to human fatigue and oversight
- **Difficult to scale:** Impractical in large crowds or continuous monitoring
- **Non-contactless:** Increases disease transmission risk during enforcement

Project Objectives

This project aims to develop a real-time face mask detection application that:

- Processes video input from webcam or file at real-time speeds
- Detects human faces using OpenCV's Haar Cascade classifier
- Classifies each detected face as "mask" or "no-mask"
- Achieves high accuracy ($> 95\%$) with emphasis on recall for "no-mask" class
- Demonstrates transfer learning effectiveness with limited training data
- Provides foundation for deployment in public health monitoring

Methodology

System Architecture

The complete pipeline consists of six sequential stages:

1. **Video Input: real-time** webcam (cv2.VideoCapture) or video file, target 30 FPS
2. **Preprocessing:** resize frames, BGR→RGB conversion, normalize [0,1]
3. **Face Detection:** OpenCV Haar Cascade parameters:
   - scaleFactor: 1.1
   - minNeighbors: 5
   - minSize: (20, 20)
4. **Classification:** MobileNetV2 model on 224×224 face crops
5. **Post-Processing:** confidence thresholding (≥ 0.80 for "mask")
6. **Output:** annotated frames with bounding boxes, labels, confidence

Transfer Learning Approach

Why Transfer Learning?

- Leverages knowledge from large source dataset (ImageNet: 1.4M images)
- Achieves high accuracy with limited labeled data (dataset: 170 images)
- Dramatically reduces training time (hours vs days/weeks from scratch)
- Reduces computational requirements
- Enables deployment on edge devices

How it works:

1. Pretrained model learns hierarchical features: edges → textures → shapes → objects
2. Features transfer to new domains with minimal fine-tuning
3. Only final classification layers retrained for target task

Training & Data

Dataset Composition

Total Images: 170

- With mask: 144 images
- Without mask: 10 images
- **Original imbalance:** 14.4:1 ratio

Train/Validation Split:

- Training: 136 images (80%)
  - Mask: 115 images
  - No-mask: 21 images (after oversampling)
- Validation: 34 images (20%)
  - Mask: 29 images
  - No-mask: 5 images

Training Configuration

Optimizer: Adam

- Learning rate: 1×10^{-4}
- β_1: 0.9, β_2: 0.999
- Epsilon: 1×10^{-7}

Loss Function: Categorical Cross-Entropy

Training Parameters:

- Batch size: 16 (optimized for 8GB RAM)
- Epochs: 15 (early stopping at epoch 9)
- Patience: 3 epochs

Data Augmentation (training only):

- Horizontal flip: 50% probability
- Rotation: $\pm 20°$
- Zoom: 20% scale variation
- Brightness/contrast: 20% variation

Class Imbalance Strategy

Challenge: 144 mask vs 10 no-mask (14:1 imbalance) biased predictions toward "mask" (no-mask recall ≈ 0.40).

Multi-strategy solution:

1. **Oversampling:** replicate no-mask images to achieve 1:1 ratio
2. **Class weighting:** assign higher loss weight to minority class
3. **Confidence thresholding:** require 0.80 confidence for "mask"
4. **Data augmentation:** increase effective training set size

Result: No-mask recall improved from 0.40 → 1.00

Results

Classification Performance

Table 1: Validation Set Metrics (34 images)

Class	Precision	Recall	F1-Score	Support
Mask	1.00	0.97	0.99	29
No-Mask	0.71	1.00	0.83	5
Weighted Avg	**0.97**	**0.98**	**0.98**	**34**

Overall Accuracy: 97%

Real-Time Performance

Hardware: MacBook Pro 2016 (CPU-only)

- CPU: 2.7 GHz Intel Core i7
- RAM: 8GB
- No GPU

Table 2: Inference Time Breakdown

Component	Time (ms)	30 FPS Budget
Face detection	50–100	33.3
Classification (per face)	30–50	33.3
Combined (single face)	80–150	66.6
Achieved Frame Rate: 25–29 FPS		

Confusion Matrix

Table 3: Confusion Matrix (Validation Set)

		Predicted	
		Mask	No-Mask
Actual	Mask	28	1
	No-Mask	0	5

Observation: Model errs on the side of caution (no missed no-mask cases).

Discussion

Key Achievements

- 97% accuracy exceeding the 95% target
- Perfect no-mask recall (1.00) for safety-critical detection
- 25–29 FPS real-time operation on CPU-only hardware
- Effective small-data learning via transfer learning + imbalance handling

Limitations

Dataset: small size (170), limited diversity/demographics, mostly controlled conditions.

Model: Haar cascades fail on angled faces ($> 45°$); cannot detect incorrect mask usage; no tracking.

Practical: privacy considerations; CPU-only performance may drop in crowded scenes.

Future Work

- Replace Haar cascades with MTCNN/RetinaFace
- Expand dataset (1000+ per class), varied masks/lighting/angles
- 3-class model: correct / incorrect / no mask
- Add tracking + alerts; package for edge devices; REST API

Conclusions

Conclusions

1. Transfer learning enabled 97% accuracy with only 170 images.
2. Combined imbalance strategies achieved perfect no-mask recall.
3. Real-time speed (25–29 FPS) is feasible without a GPU.

References

Selected References

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